On Determining the Independent Point Set for Doubly Periodic Arrays and Encoding Two-Dimensional Cyclic Codes and Their Duals

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Abstract—**For the purpose of encoding two-dimensional cyclic (TDC) codes, an effective algorithm for finding the independent point set of an arbitrary module of doubly periodic (DP) arrays is proposed. In addition, a method for determining the characteristic ideal of a given set of DP arrays is exhibited. With the aid of these methods it is possible to specify the structure of the generator and check ideals of TDC codes. By applying this algorithm to nonsemisimple binary TDC codes with small areas, several optimal linear codes have been found and their previously unknown TDC structures have been exhibited.**

I. Introduction

DOUBLY periodic (DP) arrays [10] and two-dimensional cyclic (TDC) codes [7], [8] are two-dimensional generalizations of periodic sequences over finite fields and of well-known cyclic codes, respectively. TDC codes are also families of Abelian codes [1], [2], [9] (alias multidimensional cyclic codes) which show promise. But they have not yet led to any really good codes, largely because the basic theory, including the concepts of generator polynomials and encoding by feedback shift-registers, is incomplete. We have already investigated properties of DP arrays in a general fashion [10]. However, there remained an open problem of generating DP arrays, i.e., encoding TDC codes. More precisely, there has been no known method of determining the positions of information symbols for nonsemisimple TDC codes, although semisimple cases are completely treated by the zeros of ideals in the ring R of bivariate polynomials [2], [8], [9].

A nonzero bivariate polynomial $f(x, y)$ is said to have quasi-degree $\mathrm{Deg}\, f \triangleq (\kappa, \lambda)$ if $f(x, y)$ has a nonzero term $f_{\kappa,\lambda}x^{\kappa}y^{\lambda}$ but not any nonzero term $f_{i,j}x^{i}y^{j}$ such that $j > \lambda$ or both $j = \lambda$ and $i > \kappa$ hold [7]. Provided that $\mathrm{Deg}\, f_1 = (\kappa_1, \lambda_1) \geq \mathrm{Deg}\, f_2 = (\kappa_2, \lambda_2)$, i.e., $\kappa_1 \geq \kappa_2$ and $\lambda_1 \geq \lambda_2$, there exist polynomials h and g which satisfy $f_1 = f_2 h + g$. It has been shown that, if $\{f_1, \cdots, f_M\}$ is a maximal set of polynomials with minimal quasi-degree in an ideal $\mathcal{A}$, i.e., $\mathrm{Deg}\, f_k \not\geq \mathrm{Deg}\, f_l$, $k \neq l$, $1 \leq k, l \leq M$, and, for any nonzero polynomial f in $\mathcal{A}$, there exists a polynomial f_k in the set such that $\mathrm{Deg}\, f \geq \mathrm{Deg}\, f_k$, then the codewords of the TDC code whose check ideal is $\mathcal{A}$ are generated by means of the polynomials $f_1, \cdots, f_M$ and the positions $\{(i, j) \geq (0, 0) \mid (i, j) \not\geq \mathrm{Deg}\, f_k,\ 1 \leq k \leq M\}$ are those of information symbols [7].

In this paper we present an algorithm which, starting from a given basis $\{f_1, \cdots, f_M\}$ of an arbitrary ideal $\mathcal{A}$, yields the normal basis consisting of the polynomials with minimal quasi-degree in $\mathcal{A}$. The algorithm contains two major routines. In the first normalization routine we replace f_k, $1 \leq k \leq M$, by g such that $f_k = f_l h + g$ for some nonzero h so long as $\mathrm{Deg}\, f_k \geq \mathrm{Deg}\, f_l$, $k \neq l$; $1 \leq l \leq M$. In the second routine we check the consistency of the normalized system of polynomials $f_1, \cdots, f_M$. If they pass the test, they are the normal basis of $\mathcal{A}$. Conversely, if they do not pass the test a certain polynomial f in $\mathcal{A}$ satisfying $\mathrm{Deg}\, f \not\geq \mathrm{Deg}\, f_k$, $1 \leq k \leq M$, is produced during this routine. The polynomials $f_1, \cdots, f_M$ and f are then normalized again. Continuing in this manner, it is possible to obtain the normal basis of $\mathcal{A}$ in a finite number of steps. The algorithm can also take the place of the classical "Noether's condition" [12] under which it is possible (but not efficient) to judge whether a given polynomial is contained in $\mathcal{A}$.

Making use of our algorithm, we have explored binary TDC codes with even area, which are non-semisimple. Any non-semisimple TDC code is a direct sum of quasi-irreducible codes whose check ideals are primary in the ring R. After searching among quasi-irreducible TDC codes with small areas by means of computing machinery (Tokyo University HITAC 8800 · 8700 system), we have found several optimal linear codes and their previously unknown TDC structures.

The outline of this paper is as follows. Section II is devoted to preliminaries on properties of DP arrays and TDC codes. Section III describes the normalization algorithm which yields the independent point (IP) set of an arbitrary module of DP arrays, i.e., the positions of information symbols for TDC codes. For convenience in referring to a previous paper [10], we frequently use the terminology of DP arrays which can be immediately translated into that of TDC codes. Section IV describes a

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method of determining the characteristic ideal for a given set of DP arrays. The problem of Section IV is somewhat dual to that of Section III. In Section V the result obtained by applying the normalization algorithm to binary non-semisimple codes is exhibited. Conclusions appear in Section VI.

II. Preliminaries on Doubly Periodic Arrays and Two-Dimensional Cyclic Codes

An array $u = (u_{i,j})$ over a finite field $F \triangleq \mathrm{GF}(q)$ is a mapping from the Cartesian product $Z^2 \triangleq Z \times Z$ of the ring Z of integers to the field F, where the value $u(\boldsymbol{i}) = u_{i,j} \in F$ of the array u on $\boldsymbol{i} = (i, j) \in Z^2$ is said to be the (i, j) component of u. A doubly periodic (DP) array $u = (u_{i,j})$ is defined to be an array satisfying the following relation

$$u(\boldsymbol{i}) = u(\boldsymbol{i} + \boldsymbol{l}_1) = u(\boldsymbol{i} + \boldsymbol{l}_2)$$

for all $\boldsymbol{i} \in Z^2$ and for some $\boldsymbol{l}_1, \boldsymbol{l}_2 \in Z^2$ which are linearly independent over Z, $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$ being called the period $\cdot$ vectors of u. A DP array u has a period $\cdot$ vector in an arbitrary rational direction $\boldsymbol{k} \in Z^2$. The period $\cdot$ vector with the minimum length in a direction is called a fundamental period (FP) vector. The partial x-period $\mathrm{per}_x(u)$ is the length of the FP vector in the direction of $\boldsymbol{i}_x \triangleq (1, 0)$ and the partial y-period $\mathrm{per}_y(u)$ is that of the FP vector in the direction of $\boldsymbol{i}_y \triangleq (0, 1)$.

The sum $u + v$ of two arrays u, v and scalar multiplication cu of an array u by a constant $c \in F$ are defined in the ordinary way. Then the set W of all DP arrays forms an F-module, i.e., a linear space over the field F. The identity element of the module W is the zero array (0) whose components all vanish. For two F-submodules U and V of W, $U + V$ is the submodule of W which is the set of arrays $u + v$ for $u \in U$, $v \in V$.

For a finite subset Γ of Z^2 and $f_{I,J} \in F$, $(I, J) \in \Gamma$,

$$f(x, y) = \sum_{(I,J)\in\Gamma} f_{I,J} x^I y^J$$

is defined to be a linear operator from the set of arrays to itself. Operating f on an array $u = (u_{i,j})$ yields the array $v = f(x, y)u$ whose (i, j) component is

$$v_{i,j} = \sum_{(I,J)\in\Gamma} f_{I,J} u_{i+I,j+J}, \qquad (i, j) \in Z^2.$$

These operators form a ring $\tilde{R}$ which is isomorphic to the extension ring $R[x^{-1}, y^{-1}]$ which is obtained by adjoining x^{-1}, y^{-1} to the ring $R \triangleq F[x, y]$ of bivariate polynomials with coefficients in F. Operators x, y and x^{-1}, y^{-1} are forward and backward shifts. A linear recurring (LR) relation

$$\sum_{(I,J)\in\Gamma} f_{I,J} u_{i+I,j+J} = 0, \qquad (i, j) \in Z^2,$$

which an array u satisfies, can be written in the form

$$f(x, y)u = (0).$$

The set of all LR arrays u which satisfy a system of LR relations

$$f^{(k)}(x, y)u = (0), \qquad 1 \le k \le M, \tag{1}$$

coincides with the set $G(\mathfrak{A})$ of arrays u such that, for any element $f(x, y)$ of the ideal $\mathfrak{A} = (f^{(1)}, \cdots, f^{(M)})$ generated by $f^{(1)}, \cdots, f^{(M)}$ in the ring $\tilde{R}$, $f(x, y)u = (0)$. $G(\mathfrak{A})$ is an $\tilde{R}$-module, i.e., a module which is closed under operations by elements of $\tilde{R}$.

If $\mathfrak{A}$ and $\mathfrak{L}$ are ideals in $\tilde{R}$ such that $\mathfrak{A} \supset \mathfrak{L}$, then $G(\mathfrak{A}) \subset G(\mathfrak{L})$. In particular, if $\mathfrak{A} \supset \mathfrak{J}_{m,n} \triangleq (x^m - 1, y^n - 1)$, then $G(\mathfrak{A}) \subset W_{m,n} \triangleq G(\mathfrak{J}_{m,n})$, i.e., all arrays in $G(\mathfrak{A})$ are DP arrays with the partial periods such that

$$\mathrm{per}_x(u) \mid m, \mathrm{per}_y(u) \mid n.$$

It is true that $\mathfrak{A} \supset \mathfrak{J}_{m,n}$ for certain integers m, n if and only if the algebraic set $V(\mathfrak{A})$ is finite, where $V(\mathfrak{A})$ is the totality of all zeros common to the polynomials contained in $\mathfrak{A}$.

Provided that $\mathfrak{A} \supset \mathfrak{J}_{m,n}$, there is a one-to-one correspondence between $G(\mathfrak{A})$ and a two-dimensional cyclic code (TDC) C having code area $m \times n$, where each codeword of C is an $m \times n$-rectangle part of a DP array u in $G(\mathfrak{A})$, i.e., an $m \times n$-matrix $\boldsymbol{U} = \|u_{i,j}\|$ with the subscripts $i = 0, 1, \cdots, m - 1$; $j = 0, 1, \cdots, n - 1$. Thus $G(\mathfrak{A})$ is $\tilde{R}$-isomorphic to C, where a shift of an array corresponds to a cyclic shift of a codeword. A TDC code C with area $m \times n$ is also defined to be an ideal in the residue class ring $R_{m,n} \triangleq \tilde{R}/\mathfrak{J}_{m,n}$ of $\tilde{R}$ modulo $\mathfrak{J}_{m,n}$, where each codeword $\boldsymbol{U} = \|u_{i,j}\|$ is represented as a polynomial

$$u(x, y) = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} u_{i,j} x^i y^j \pmod{\mathfrak{J}_{m,n}}.$$

Thus $C = \mathfrak{L}/\mathfrak{J}_{m,n}$ for a certain ideal $\mathfrak{L} \supset \mathfrak{J}_{m,n}$ in $\tilde{R}$. We call $\mathfrak{L}$(or $\mathfrak{L}/\mathfrak{J}_{m,n}$) the generator ideal of the code C. On the other hand, it is valid that the product of $\mathfrak{L}$ and the reciprocal ideal $\bar{\mathfrak{A}} = \{f(x^{-1}, y^{-1}) \mid f(x, y) \in \mathfrak{A}\}$ of $\mathfrak{A}$ vanishes mod $\mathfrak{J}_{m,n}$, i.e.,

$$\mathfrak{L}\bar{\mathfrak{A}} = 0 \mod \mathfrak{J}_{m,n}. \tag{2}$$

We call $\mathfrak{A}$ (or $\mathfrak{A}/\mathfrak{J}_{m,n}$) the check ideal of the code C and denote C by $C_{\mathfrak{A}}$. From (2) it follows immediately that $\mathfrak{L}$ is the check ideal of the TDC code $C' \triangleq \mathfrak{A}/\mathfrak{J}_{m,n}$ which is the 'dual code' of C, and that, for any codeword $u(x, y) = \sum_{i=0}^{m-1}\sum_{j=0}^{n-1} u_{i,j} x^i y^j$ of C and any codeword $v(x, y) = \sum_{i=0}^{m-1}\sum_{j=0}^{n-1} v_{i,j} x^i y^j$ of C', their inner product

$$(u, v) \triangleq \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} u_{i,j} v_{i,j} \quad (\in F)$$

vanishes, i.e., C and C' are orthogonal subspaces in the linear space $R_{m,n}$.

By virtue of the Lasker–Noether decomposition theorem [12], every ideal $\mathfrak{A}$ in $\tilde{R}$ has an irredundant representation as the intersection of a finite number of greatest primary components (ideals):

$$\mathfrak{A} = \bigcap_{k=1}^{M} \mathfrak{Q}_k,$$

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where $V(\mathcal{A}) = \cup_{k=1}^{M} V(\mathcal{Q}_k)$. In particular, for $\mathcal{A} \supset \mathcal{J}_{m,n}$, these primary ideals $\mathcal{Q}_k$, which belongs to distinct (maximal) prime ideals $\mathcal{P}_k$, $1 \le k \le M$, are relatively prime to one another where, for a certain integer σ_k, $\mathcal{P}_k \supset \mathcal{Q}_k \supset \mathcal{P}_k^{\sigma_k}$ and $V(\mathcal{Q}_k) = V(\mathcal{P}_k)$ hold.[1] Therefore $\mathcal{A}$ can be rewritten in the form

$$\mathcal{A} = \prod_{k=1}^{M} \mathcal{Q}_k, \tag{3}$$

and $G(\mathcal{A})$ is decomposed into the direct sum

$$G(\mathcal{A}) = G(\mathcal{Q}_1) \dotplus \cdots \dotplus G(\mathcal{Q}_M).$$

Correspondingly any TDC code $C_{\mathcal{A}}$ is the direct sum of $C_{\mathcal{Q}_k}$, $1 \le k \le M$. If m, n are not divisible by the characteristic p of F, a TDC code $C_{\mathcal{A}}$ with area $m \times n$ is called a 'semisimple' code and can be decomposed into a direct sum

$$C_{\mathcal{A}} = C_{\mathcal{P}_1} \dotplus \cdots \dotplus C_{\mathcal{P}_M}$$

of irreducible codes $C_{\mathcal{P}_k}$ defined by prime ideals $\mathcal{P}_k$, $1 \le k \le M$ [2], [9]. An irreducible TDC code $C_{\mathcal{P}}$ is minimal in $R_{m,n}$ except for the trivial code $\{0\}$. For integers m, n, either of which is divisible by p, a TDC code $C_{\mathcal{A}}$ with area $m \times n$ is non-semisimple and is a direct sum of quasi-irreducible codes $C_{\mathcal{Q}_k}$ defined by primary ideals $\mathcal{Q}_k$, $1 \le k \le M$.

For an integral unimodular matrix $\boldsymbol{S} = \begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$, where $\det(\boldsymbol{S}) = 1$ or -1, the array $u' = \phi_{\boldsymbol{S}}(u)$ rearranged from an array u by means of $\boldsymbol{S}$ is defined by

$$u'(\boldsymbol{i}) = u(\boldsymbol{i}\boldsymbol{S}), \qquad \boldsymbol{i} \in Z^2.$$

Furthermore, the unimodular transformation $\boldsymbol{S}$ induces the following ring-automorphism $\psi_{\boldsymbol{S}}$ of $\tilde{R}$:

$$\psi_{\boldsymbol{S}}(f(x, y)) \triangleq f(x^a y^b, x^c y^d), \qquad f(x, y) \in \tilde{R}.$$

The set $\phi_{\boldsymbol{S}}(G(\mathcal{A}))$ of arrays $\phi_{\boldsymbol{S}}(u)$ rearranged from arrays u in $G(\mathcal{A})$ coincides with $G(\psi_{\boldsymbol{S}}^{-1}(\mathcal{A}))$. Hence, if $\mathcal{A} \supset \mathcal{J}_{m,n}$, $\mathcal{A}' = \psi_{\boldsymbol{S}}^{-1}(\mathcal{A}) \supset \mathcal{J}_{m',n'}$ and $mn = m'n'$ for integers m, n, m', n', then the code $C_{\mathcal{A}}$ with area $m \times n$ is equivalent (as an error-correcting code) to the code $C_{\mathcal{A}'}$. For example, if m and n are relatively prime to each other, any TDC code with area $m \times n$ is equivalent to a cyclic code with length mn. Furthermore, any irreducible TDC code with area $m \times n$ is equivalent to an irreducible cyclic code (or its repetition code) even if m and n are not relatively prime to each other.

Punctuation and shortening of TDC codes can be defined in the same fashion as cyclic codes. For instance, deleting k columns (or rows) from each codeword (matrix) of a TDC code with area $m \times n$ yields a quasi-cyclic code with length $m(n-k)$ (or $n(m-k)$).

III. Determining Independent Point Sets of Doubly Periodic Arrays

For an ideal $\mathcal{A} \supset \mathcal{J}_{m,n}$, $G(\mathcal{A})$ is a linear space over F having the dimension

$$\dim G(\mathcal{A}) = \dim_F \tilde{R}/\mathcal{A},$$

where the right side is the dimension of the residue class ring $\tilde{R}/\mathcal{A}$ as an F-module [10]. Let $K = \dim G(\mathcal{A})$, then there exists a certain subset Γ of Z^2 such that $|\Gamma| = K$ and a unique array u in $G(\mathcal{A})$ is determined by the components $u_{i,j}$, $(i, j) \in \Gamma$ which can be given *a priori* by an arbitrary set of elements in F. Such a subset Γ of Z^2 is said to be an independent point (IP) set $\Delta(\mathcal{A})$ of $G(\mathcal{A})$ (or $\mathcal{A}$) which coincides with the positions of information symbols of the corresponding TDC code $C_{\mathcal{A}}$. $\Delta(\mathcal{A})$ can be characterized as follows.

Proposition [10, thm. 10]: A subset Γ of Z^2 is an IP set $\Delta(\mathcal{A})$ of an ideal $\mathcal{A} \supset \mathcal{J}_{m,n}$ if and only if the following conditions a) and b) are valid.

a) $\mathcal{A} \cap \boldsymbol{P}(\Gamma) = \phi$, where

$$\boldsymbol{P}(\Gamma) \triangleq \left\{ f(x, y) = \sum_{(I,J)\in\Gamma} f_{I,J} x^I y^J (\neq 0) \,|\, f_{I,J} \in F \right\} \subset \tilde{R}; \tag{4}$$

b) Γ is a maximal subset of Z^2 such that a) is valid, or equivalently, $|\Gamma| = \dim G(\mathcal{A})$.

Then it is obvious that,

c) for $(k, l) \notin \Delta(\mathcal{A})$, there is a unique element

$$g(x, y) = x^k y^l - g^{(k,l)}(x, y) \in \mathcal{A},$$
$$g^{(k,l)}(x, y) \in \boldsymbol{P}(\Delta(\mathcal{A})). \tag{5}$$

A special kind of IP set $\Delta(\mathcal{A})$ in $Z_+^2 \triangleq \{(i, j) \,|\, i, j \ge 0\} \subset Z^2$ can be introduced as follows. Z_+^2 is a partially ordered set with respect to the relation that $(i, j) \ge (k, l)$ if and only if $i \ge k$ and $j \ge l$. Furthermore, Z_+^2 can be also totally ordered by a kind of lexicographic order $\overrightarrow{\ge}$, where $(i, j) \overrightarrow{\ge} (k, l)$ if and only if $j > l$ or both $j = l$ and $i \ge k$. $(i, j) \ge (k, l)$ implies that $(i, j) \overrightarrow{\ge} (k, l)$, but the converse does not necessarily hold. For a nonzero polynomial

$$f(x, y) = \sum_{(I,J)} f_{I,J} x^I y^J \in R = F[x, y],$$

let $\Gamma(f) \triangleq \{(I, J) \,|\, f_{I,J} \ne 0\} \subset Z_+^2$. The element (i, j) which is the maximum with respect to the order $\overrightarrow{\ge}$ in the finite subset $\Gamma(f)$ of Z_+^2 is called the quasi-degree of f and is denoted by $\mathrm{Deg}\, f$ [7]. For an ideal $\mathcal{A}$ in $\tilde{R}$, we define a system of polynomials $B = \{f^{(1)}, \cdots, f^{(M)}\}$ to be consistent if and only if B is a maximal set of polynomials with minimal quasi-degree in the ideal $\mathcal{A} \cap R$ of R, i.e., $\mathrm{Deg}\, f^{(k)} \not\ge \mathrm{Deg}\, f^{(l)}$, $k \ne l$, $1 \le k, l \le M$, and for any nonzero polynomial f in $\mathcal{A} \cap R$, there exists at least one polynomial $f^{(k)} \in B$ such that $\mathrm{Deg}\, f \ge \mathrm{Deg}\, f^{(k)}$. Then B is a basis of $\mathcal{A}$. Obviously any subset of a consistent system B is also consistent (with respect to a subideal of $\mathcal{A}$). Let $B = \{f^{(1)}, \cdots, f^{(M)}\}$ be a consistent basis of $\mathcal{A} \supset \mathcal{J}_{m,n}$ in $\tilde{R}$, and set

$$\Delta = \bigcap_{k=1}^{M} \Delta(f^{(k)}), \tag{6}$$

where $\Delta(f^{(k)}) \triangleq \{(i, j) \in Z_+^2 \,|\, (i, j) \not\ge \mathrm{Deg}\, f^{(k)}\}$, $1 \le k \le M$. It is easy to show that Δ satisfies conditions a) and b) in the proposition. Therefore it is an IP set $\Delta(\mathcal{A})$ for $\mathcal{A}$ [7], which we call the "normal IP set" of $G(\mathcal{A})$.

[1] For such ideal theory notation, refer to, e.g., [12].

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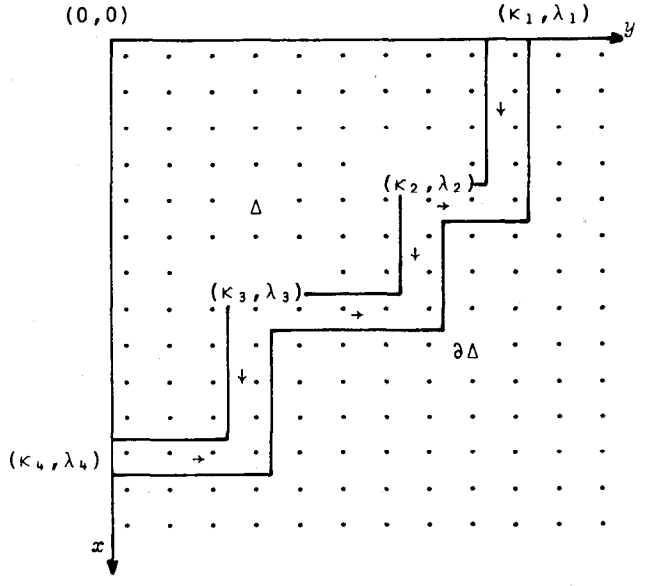


Fig. 1. Example of normal IP set and its periphery.

The set Δ is a subset of the $m \times n$-rectangle $Z^2_{m,n} \triangleq \{(i, j) \mid 0 \le i \le m-1, 0 \le j \le n-1\}$. We may renumber $f^{(1)}, \cdots, f^{(M)}$ such that $\mathrm{Deg}\, f^{(1)} \to \mathrm{Deg}\, f^{(2)} \to \cdots \to \mathrm{Deg}\, f^{(M)}$, where $(i, j) \to (k, l)$ means that $(i, j) \overrightarrow{=} (k, l)$ and $(i, j) \neq (k, l)$. Then, for $\mathrm{Deg}\, f^{(s)} = (\kappa_s, \lambda_s)$, $1 \le s \le M$, it holds that

$$\kappa_1 = 0 < \kappa_2 < \cdots < \kappa_M \le m;$$
$$n \ge \lambda_1 > \lambda_2 > \cdots > \lambda_M = 0. \tag{7}$$

Thus Δ is of echelon form (Fig. 1) and is rewritten as

$$\Delta = \bigcup_{s=1}^{M-1} Z^2_{\kappa_{s+1}, \lambda_s}. \tag{8}$$

As in (5), there exist unique polynomials $g^{(s)}(x, y) = x^{\kappa_s} y^{\lambda_s} - g^{(\kappa_s, \lambda_s)}(x, y) \in \mathcal{A}$ such that $g^{(\kappa_s, \lambda_s)}(x, y) \in \boldsymbol{P}(\Delta)$, $1 \le s \le M$. Obviously they also constitute a consistent basis of $\mathcal{A}$, which we call the "normal basis" of $\mathcal{A}$. For the TDC code $C_{\mathcal{A}}$ corresponding $G(\mathcal{A})$, these polynomials are called "check polynomials". By means of these, we can calculate consecutively $g^{(i,j)}(x, y)$, $(i, j) \notin \Delta$, and the components of the array $u = (u_{i,j}) \in G(\mathcal{A})$ for a given set of values $u_{i,j}$, $(i, j) \in \Delta$. First we get $g^{(\kappa_M, j)} \in \boldsymbol{P}(\Delta)$ and $u_{\kappa_M, j}$, $0 \le j \le \lambda_{M-1} - 1$, by means of $g^{(M)}$. Second, we get $g^{(i, \lambda_{M-1})}$ and $u_{i, \lambda_{M-1}}$, $\kappa_{M-1} \le i \le \kappa_M$, by means of $g^{(M-1)}$, $g^{(M)}$ and the above-obtained $g^{(\kappa_M, j)}$ and $u_{\kappa_M, j}$, $0 \le j \le \lambda_{M-1} - 1$. Continuing in this way, we can get the polynomials $g^{(i,j)}$ and the values $u_{i,j}$, $(i, j) \in \partial\Delta$, where for any finite subset Γ of Z^2_+, the periphery $\partial\Gamma$ of Γ (Fig. 1) is defined as

$$\partial\Gamma \triangleq \{\boldsymbol{l} + \boldsymbol{i}_x, \boldsymbol{l} + \boldsymbol{i}_y \mid \boldsymbol{l} \in \Gamma\} - \Gamma.$$

Quite similarly, we may proceed to calculate consecutively other $g^{(i,j)}$ and $u_{i,j}$, $0 \le i < m$, $0 \le j < n$, periphery after periphery.

In the subsequent paragraphs, we propose the normalization algorithm which, starting from a given basis $\{f^{(1)}, \cdots, f^{(M)}\}$ of an arbitrary ideal $\mathcal{A} \supset \mathcal{I}_{m,n}$, yields both the normal basis of the ideal $\mathcal{A}$ and the normal IP set $\Delta(\mathcal{A})$.

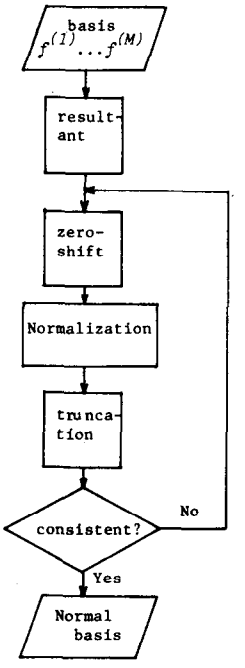


Fig. 2. Flowchart of normalization algorithm.

The flowchart of the algorithm is shown in Fig. 2. We may assume that $f^{(1)}, \cdots, f^{(M)} \in \mathcal{A} \cap R$ and that both their lowest x- and y- degrees are 0, i.e., each polynomial $f^{(k)}$ has nonzero terms $c_k x^i$ and $d_k y^j$, $c_k, d_k \in F$, $1 \le k \le M$; otherwise we have only to multiply $f^{(k)}$ by a power $x^i y^j$. We call the latter procedure "zero-shift".

If there does not exist a univariate polynomial $f(x)$ with the single indeterminate x in the set $B = \{f^{(1)}, \cdots, f^{(M)}\}$, by eliminating y from a system of algebraic equations

$$f^{(k)}(x, y) = 0, \quad f^{(l)}(x, y) = 0$$

for two polynomials $f^{(k)}$ and $f^{(l)}$ in the set B, we can obtain such a polynomial $f(x) \in \mathcal{A} = (f^{(1)}, \cdots, f^{(M)})$, which is called the 'resultant' of $f^{(k)}$ and $f^{(l)}$. In order to find the resultant of the two polynomials, we may make use of the euclidean algorithm for multivariate polynomials [4].

Provided that $\mathrm{Deg}\, f_1 \ge \mathrm{Deg}\, f_2$, there exists a pair of unique polynomials $h(\neq 0)$, $g \in R$ which satisfy $f_1 = hf_2 + g$ and $\mathrm{Deg}\, f_2 \not\le \mathrm{Deg}\, g (< \mathrm{Deg}\, f_1)$. We call this procedure "division" of f_1 by f_2, where g is the residue of $f_1 \bmod f_2$ and h is the quotient. Now the normalization routine is introduced as follows.

Algorithm 1—Normalization Procedure:

i) For a set of polynomials $B = \{f^{(1)}, \cdots, f^{(M)}\}$ in which at least one polynomial $f(x)$ is contained, if there exists a pair of polynomials $f^{(k)}$ and $f^{(l)}$ in B such that $\mathrm{Deg}\, f^{(k)} \ge \mathrm{Deg}\, f^{(l)}$, then divide $f^{(k)}$ by $f^{(l)}$ and substitute $f^{(k)}$ by the residue of $f^{(k)} \bmod f^{(l)}$, throwing away zero out of B.

ii) If for any two polynomials $f^{(k)}$ and $f^{(l)}$, $k \neq l$, in B, $\mathrm{Deg}\, f^{(k)} \not\ge \mathrm{Deg}\, f^{(l)}$, then stop.

After the normalization procedure has been completed, by renumbering if necessary, we have a new set of polynomials $f^{(k)}$ with quasi-degree (κ_k, λ_k), $1 \le k \le M$, such

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that

$$(0 \le)\kappa_1 < \kappa_2 < \cdots < \kappa_M;$$
$$\lambda_1 > \lambda_2 > \cdots > \lambda_M = 0, \tag{9}$$

where the cardinality M may be different from the initial one. Thus we have a basis $B = \{f^{(1)}, \cdots, f^{(M)}\}$ of $\mathcal{Q}$ and a set

$$\Delta = \Delta_{f^{(1)}, \cdots, f^{(M)}} = \bigcap_{k=1}^{M} \Delta(f^{(k)}) \subset Z_+^2 \tag{10}$$

of echelon form as in (8). However, B is not necessarily consistent and Δ is not necessarily an IP set of $\mathcal{Q}$.

Before inquiring about the consistency of B, we successively find, for $k = M, M-1, \cdots, 1$, the polynomials $\phi_k(x, y) \in \boldsymbol{P}(\Delta)$ which satisfy

$$f^{(k)} \equiv x^{\kappa_k} y^{\lambda_k} - \phi_k(x, y) \bmod \left(f^{(k+1)}, \cdots, f^{(M)}\right), \tag{11}$$

under the admissible assumption that each $f^{(k)}$ is monic, i.e., the coefficient of the term with degree (κ_k, λ_k) is one where the right-side polynomials in (11) are called "truncated" to within Δ. For this purpose we introduce the following "quasi-division" procedure of a polynomial f by any system of polynomials $f^{(1)}, \cdots, f^{(M)}$ (in this order).

Algorithm 2—Quasi-Division:

i) For the smallest number k such that f has at least one nonzero term $f_{i,j}x^i y^j$ with $(i, j) \ge (\kappa, \lambda) \triangleq \operatorname{Deg} f^{(k)}$, divide $f^{(\kappa,\lambda)} = \sum_{(i,j)\ge(\kappa,\lambda)} f_{i,j}x^i y^j$ by $f^{(k)}$. For the residue $g^{(\kappa,\lambda)}$ of $f^{(\kappa,\lambda)} \bmod f^{(k)}$, let $g = f - f^{(\kappa,\lambda)} + g^{(\kappa,\lambda)}$.

ii) If, for $\Delta = \cap_{k=1}^{M} \Delta(f^{(k)})$, $g \notin \boldsymbol{P}(\Delta)$, then replace f by g and go to i); otherwise stop.

We call the polynomial g which has been obtained finally by the name of the "quasi-residue" $\operatorname{res}_{f^{(1)}, \cdots, f^{(M)}}(f)$ of the initial $f \bmod f^{(1)}, \cdots, f^{(M)}$. It is valid that $\operatorname{Deg} g \overset{\leftarrow}{=} \operatorname{Deg} f$, $g \in \boldsymbol{P}(\Delta)$, and $f \equiv g \bmod (f^{(1)}, \cdots, f^{(M)})$. It should be noticed that the quasi-residue of $f \bmod f^{(1)}, \cdots, f^{(M)}$ depends on the ordering of $f^{(1)}, \cdots, f^{(M)}$ unless these polynomials are consistent. Thus we can put

$$\phi_k(x, y) = \operatorname{res}_{f^{(k+1)}, \cdots, f^{(M)}}\left(x^{\kappa_k} y^{\lambda_k} - f^{(k)}\right)$$

in (11), assuming each $f^{(k)}$ to be monic.

Whether the above-obtained polynomials $f^{(1)}, \cdots, f^{(M)}$ are consistent is decided upon by the following lemma, since $\{f^{(M)}\}$ is consistent.

Lemma 1: Let $f^{(k+1)}, \cdots, f^{(M)}$ be consistent, $1 \le k < M$. $f^{(k)}, f^{(k+1)}, \cdots, f^{(M)}$ are consistent if and only if there exists a unique polynomial $g(x, y) \in \boldsymbol{P}(\Delta_{f^{(k)}f^{(k+1)}\cdots f^{(M)}})$ such that $\operatorname{Deg} g \leftarrow \operatorname{Deg} f^{(k)}$ and

$$x^{\kappa_{k+l}} y^{\lambda_k} \equiv g(x, y) \bmod \left(f^{(k)}, f^{(k+1)}, \cdots, f^{(M)}\right). \tag{12}$$

Practically, $g(x, y)$ in (12) can be obtained in the following two ways. Therefore we have only to ask whether the same $g(x, y)$ is obtained in each way.

Algorithm 3—Check of Consistency: Let $\Phi^{(0)} = \phi_k$, $\Psi^{(0)} = \phi_{k+1}$. Calculate successively:

a) for $i = 1, \cdots, \kappa(\triangleq \kappa_{k+1} - \kappa_k > 0)$,

$$\Phi^{(i)} = \operatorname{res}_{f^{(k)}f^{(k+1)}\cdots f^{(M)}}\left(x\Phi^{(i-1)}\right), \tag{13}$$

b) for $j = 1, \cdots, \lambda(\triangleq \lambda_k - \lambda_{k+1} > 0)$,

$$\Psi^{(j)} = \operatorname{res}_{f^{(k)}f^{(k+1)}\cdots f^{(M)}}\left(y\Psi^{(j-1)}\right). \tag{14}$$

Then, $f^{(k)}, f^{(k+1)}, \cdots, f^{(M)}$ are consistent so long as $\Phi^{(\kappa)} = \Psi^{(\lambda)}$.

If $\Phi^{(\kappa)} \ne \Psi^{(\lambda)}$, adding $f = \Phi^{(\kappa)} - \Psi^{(\lambda)} \in \boldsymbol{P}(\Delta) \cap \mathcal{Q}$ to the set B, we return to the normalization routine. In the latter case the initial $\Delta_{f^{(1)}\cdots f^{(M)}}$ is replaced by its proper subset. After repeating the above procedures, we never fail to obtain a consistent system B of polynomials and the normal IP set $\Delta(\mathcal{Q})$ in a finite number of steps. The computational complexity is about $O(d^4)$ for the highest degree d of polynomials in the initial B.

Example 1: Let $\mathcal{Q} = (y^2 + 1, (x+1)y + x^3 + 1, x^4 + 1)$, where $F = \mathrm{GF}(2)$. $f^{(3)} = x^4 + 1$ and $f^{(2)} = (x+1)y + x^3 + 1$ are consistent. For $f^{(3)}$, $f^{(2)}$, and $f^{(1)} = y^2 + 1$, let $\Phi^{(0)} = 1$ $(\equiv y^2 \bmod \mathcal{Q})$, $\Psi^{(0)} = y + x^3 + 1$ $(\equiv xy \bmod \mathcal{Q})$. It is immediate to show that $\Phi^{(1)} = \operatorname{res}(x\Phi^{(0)}) = x$, $\Psi^{(1)} = \operatorname{res}(y\Psi^{(0)}) = x^3 + x^2 + 1$, and that $f = \Phi^{(1)} - \Psi^{(1)} = x^3 + x^2 + x + 1 \in \boldsymbol{P}(\Delta) \cap \mathcal{Q}$. Finally we get the normal basis $(y^2 + 1, (x+1)(y+x), (x+1)^3)$ of $\mathcal{Q}$.

Remark 1: We can easily make use of the normalization algorithm in order to determine the sum $\mathcal{Q} + \mathcal{L}$ and the product $\mathcal{Q} \cdot \mathcal{L}$ for given ideals $\mathcal{Q} = (f_1, \cdots, f_M)$ and $\mathcal{L} = (g_1, \cdots, g_L)$.

Remark 2: According to Noether's condition [12], it is possible to determine whether a polynomial $f(x, y)$ is contained in a given ideal $\mathcal{Q} = (f_1, \cdots, f_M)$. Hence, we can also explore polynomials with minimal quasi-degree in $\mathcal{Q}$. But this method is not efficient since it is necessary to solve directly a large system of linear equations with respect to the coefficients of the polynomials [6]. Our normalization algorithm takes full advantage of the sparsity of the matrix under investigation. Furthermore, once the normal basis of $\mathcal{Q}$ is obtained, it is easy to judge by quasi-division whether a given polynomial f is contained in $\mathcal{Q}$.

IV. Determining Independent Point Sets of Characteristic Ideals

For a finite nonempty subset U of W, the ideal

$$\mathcal{Q}(U) \triangleq \left\{f(x, y) \in \tilde{R} \mid f(x, y)u = (0), u \in U\right\}$$

is said to be the "characteristic ideal" of U [10]. In this section, we present an algorithm to determine the normal IP set $\Delta(\mathcal{Q})$ for the characteristic ideal $\mathcal{Q} = \mathcal{Q}(U)$ of a given set of DP arrays $U = \{u^{(1)}, \cdots, u^{(L)}\}$. In the case where $U = G(\mathcal{Q})$, it gives a construction method for the two-dimensional feedback shift-register with minimum number of storage devices that generates the codewords of the TDC code $C_{\mathcal{Q}}$.

Let $U \subset W_{m,n}$. We begin by determining a subset Γ_0 of the $m \times n$-rectangle $Z_{m,n}^2$ such that, for any $(i, j) \notin \Gamma_0$, the shifted arrays $x^i y^j u^{(l)}$, $1 \le l \le L$, are linearly dependent on the set of arrays

$$U(\Gamma_0) \triangleq \left\{x^i y^j u^{(l)} \mid (i, j) \in \Gamma_0, 1 \le l \le L\right\}.$$

For example, let Γ_0 be $\cup_{l=1}^{L} \Delta(\mathcal{Q}_l)$, where each $\mathcal{Q}_l \triangleq \mathcal{Q}(\{u^{(l)}\})$ is the characteristic ideal of a single DP array $u^{(l)}$ and its IP set $\Delta(\mathcal{Q}_l)$ can be obtained by using the

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method described in [10]. Let $\{v^{(1)}, \cdots, v^{(K)}\}$ be a maximal subset of linearly independent arrays in $U(\Gamma_0)$, where $K = \dim G(\mathcal{A}) \geq |\Gamma_0|$. Now we are ready to execute the following algorithm.

Algorithm 4:

i) For $k = 0, 1, \cdots$, let Γ_{k+1} be $\Gamma_k \cup \{\boldsymbol{i}_k\}$, where $\boldsymbol{i}_k = (i_k, j_k) \in \partial\Gamma_k$ is chosen so that the following condition is satisfied. For $r_k = |\Gamma_k|$ and

$$x^i y^j u^{(\lambda)} = \sum_{\kappa=1}^{K} \beta_{i,j}^{(\kappa,\lambda)} v^{(\kappa)}, \qquad (i, j) \in \Gamma_k,\ 1 \leq \lambda \leq L,$$

$$x^{i_k} y^{j_k} u^{(\lambda)} = \sum_{\kappa=1}^{K} \alpha_{i_k,j_k}^{(\kappa,\lambda)} v^{(\kappa)}, \qquad 1 \leq \lambda \leq L, \tag{15}$$

the rank of the $(r_k + 1) \times KL$ matrix

$$\boldsymbol{A} = \underset{\leftarrow\ r_k + 1\ \rightarrow}{\| \beta_{i,j}^{(\kappa,\lambda)}\ \alpha_{i_k,j_k}^{(\kappa,\lambda)} \|}\ \begin{matrix}\uparrow \\ KL \\ \downarrow\end{matrix}$$

is greater than that of the $r_k \times KL$ matrix

$$\boldsymbol{B} = \underset{\leftarrow\ r_k\ \rightarrow}{\| \beta_{i,j}^{(\kappa,\lambda)} \|}\ \begin{matrix}\uparrow \\ KL \\ \downarrow\end{matrix}$$

(by one).

ii) If there exists no such point $\boldsymbol{i}_k \in \partial\Gamma_k$ for some k, then stop with $\Delta(\mathcal{A}) = \Gamma_k$.

Provided that $|\Gamma_0| = K$, $\Gamma_0 = \Delta(\mathcal{A})$ holds, which is obvious for $L = 1$.

Now that the IP set $\Delta = \Delta(\mathcal{A})$ is obtained, we can find the polynomials $h^{(i,j)}(x, y) = x^i y^j - \Sigma(I, J) \in \Delta^{\gamma} I$, $J^{x^i y^j}$ $(\in \mathcal{A})$ for $(i, j) \in \partial\Delta$ by solving the following system of linear equations derived from (15)

$$\alpha_{i,j}^{(\kappa,\lambda)} = \sum_{(I,J)\in\Delta} \beta_{I,J}^{(\kappa,\lambda)} \gamma_{I,J}, \qquad 1 \leq \kappa \leq K,\ 1 \leq \lambda \leq L,$$

which has a unique solution for $\gamma_{I,J}$, $(I, J) \in \Delta$. In particular, we can obtain $\Delta = \cup_{k=1}^{M-1} Z^2_{\kappa_{k+1},\lambda_k}$ of echelon form, in which case $h^{(\kappa_1,\lambda_1)}, \cdots, h^{(\kappa_M,\lambda_M)}$ is the normal basis of $\mathcal{A}$.

Example 2: For binary arrays $u^{(1)}, u^{(2)} \in W_{6,6}$ (Fig. 3), their characteristic ideals are $\mathcal{A}_1 = \mathcal{A}(u^{(1)}) = (x^4 + x^2 + 1, y + x^3 + x + 1)$ and $\mathcal{A}_2 = \mathcal{A}(u^{(2)}) = (x^2 + x + 1, y^2 + x^2)$, respectively. Therefore, $\Gamma_0 = \{(0,0), (1,0), (2,0), (3,0), (0,1), (1,1)\}$. For $\mathcal{A} = \mathcal{A}_1 \cap \mathcal{A}_2 = \mathcal{A}(\{u^{(1)}, u^{(2)}\})$, $K = \dim G(\mathcal{A}) = 6$; $\Delta(\mathcal{A}) = \Gamma_0$, since $v^{(1)} \triangleq u^{(1)}$, $v^{(2)} \triangleq xu^{(1)}$, $v^{(3)} \triangleq x^2 u^{(1)}$, $v^{(4)} \triangleq x^3 u^{(1)}$, $v^{(5)} \triangleq u^{(2)}$, and $v^{(6)} \triangleq xu^{(2)}$ are linearly independent. From the following matrix

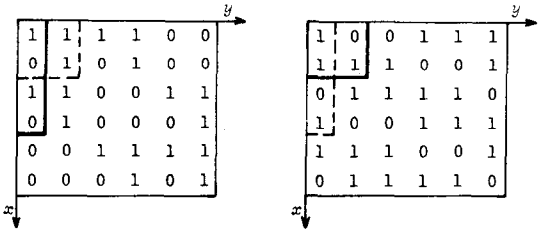


Fig. 3. Examples of binary DP arrays $\in W_{6,6}$.

we obtain $\mathcal{A} = (x^4 + x^2 + 1, (x^2 + x + 1)(y + x), y^2 + x^2)$.

Suppose that $\{f^{(1)}, \cdots, f^{(M)}\}$ constitutes a basis for the generator ideal $\mathcal{L}$ of a TDC code $C = C_{\mathcal{A}} = \mathcal{L}/\mathcal{J}_{m,n}$, where DP arrays $u^{(1)}, \cdots, u^{(M)} \in G(\mathcal{A})$ correspond to codewords $f^{(1)}, \cdots, f^{(M)}$ of C, respectively. Then the characteristic ideal of $U = \{u^{(1)}, \cdots, u^{(M)}\}$ coincides with the reciprocal $\bar{\mathcal{A}}$ to the check ideal $\mathcal{A}$ of the code C, by virtue of (2). Therefore we can obtain the normal basis of $\mathcal{A}$, i.e., the check polynomials of C, by applying Algorithm 4 to $\bar{U} = \{\bar{u}^{(1)}, \cdots, \bar{u}^{(M)}\}$, where each $\bar{u}^{(k)} = (\bar{u}_{i,j}^{(k)})$ is the reciprocal array of $u^{(k)}$, $1 \leq k \leq M$, defined as

$$\bar{u}_{i,j}^{(k)} = u_{-i,-j}^{(k)}, \qquad (i, j) \in Z^2.$$

The polynomials which constitute the normal basis for the generator ideal $\mathcal{L}$ of C are called the "generator polynomials" of C. We can find the normal basis of $\mathcal{L}$ from that of $\mathcal{A}$ as follows. In general, for an ideal $\mathcal{A} \supset \mathcal{J}_{m,n}$, the $U \triangleq \{u^{(1)}, \cdots, u^{(M)}\} \subset W_{m,n}$ satisfying the condition that the characteristic ideal $\mathcal{A}(U)$ coincides with $\mathcal{A}$ is said to be a set of "representative arrays" of the ideal $\mathcal{A}$. From the IP set $\Delta(\mathcal{A})$ of $\mathcal{A}$ we can find the following representative arrays.

Lemma 2: Let $u^{(k)} = (u_{i,j}^{(k)}) \in G(\mathcal{A})$ be defined as follows

$$u_{i,j}^{(k)} = \begin{cases} 1, & (i, j) = l_k \triangleq (\kappa_{k+1} - 1, \lambda_k - 1); \\ 0, & (i, j) \in \Delta - \{l_k\}, \end{cases} \qquad 1 \leq k \leq M - 1,$$

where $\Delta \triangleq \cup_{k=1}^{M-1} Z^2_{\kappa_{k+1},\lambda_k}$ is the normal IP set of $\mathcal{A}$. Then $\mathcal{A} = \mathcal{A}(\{u^{(1)}, \cdots, u^{(M)}\})$. It is obvious that the polynomials which correspond to the reciprocal arrays $\bar{u}^{(k)}$ of the representative arrays $u^{(k)}$, $1 \leq k \leq M$, obtained as in Lemma 2 constitute the normal basis for the generator ideal $\mathcal{L}$ of C. Thus we can find the structure of the dual code for a given TDC code, whether the latter is defined by check polynomials or generator polynomials. Furthermore, by virtue

(i, j) \ (κ, λ)	(1,1)	(2,1)	(3,1)	(4,1)	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)
(0,0)	1								1	
(1,0)		1								1
(2,0)			1						1	1
(3,0)				1					1	
(0,1)	1	1		1		1	1	1		1
(1,1)	1	1			1			1	1	1
(4,0)	1		1							1
(2,1)		1	1		1	1	1		1	
(0,2)			1						1	1

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of Algorithm 4 and Lemma 2, we have a method of determining the normal basis of an ideal $\mathfrak{A} \supset \mathfrak{J}_{m,n}$ from a given basis $f^{(1)}, \cdots, f^{(M)}$ of the ideal $\mathfrak{A}$. However, we have already presented a more direct and efficient algorithm for the same purpose in the previous section.

Remark 3: Making use of Algorithm 4, we can find the normal basis of $\mathfrak{A} \cap \mathfrak{L}$ from the representative arrays $u^{(1)}, \cdots, u^{(M)}$ of $\mathfrak{A}$ and $v^{(1)}, \cdots, v^{(L)}$ of $\mathfrak{L}$, since the whole arrays $u^{(1)}, \cdots, u^{(M)}, v^{(1)}, \cdots, v^{(L)}$ are those of $\mathfrak{A} \cap \mathfrak{L}$.

Remark 4: If the normal basis $\{f^{(1)}, \cdots, f^{(M)}\}$ of $\mathfrak{A}$ is given together with $\Delta(\mathfrak{A})$ for a principal ideal $\mathfrak{L} = (g) \subset \tilde{R}$, we can determine the quotient ideal $\mathfrak{A}:\mathfrak{L}$ in the following way, which to some extent is similar to Algorithm 4. After having calculated

$$f^{(i,j)}(x, y) = \mathrm{res}_{f^{(1)}, \cdots, f^{(M)}}\left(x^i y^j g(x, y)\right) \in \boldsymbol{P}(\Delta),$$

for each $(i, j) \in \Delta = \Delta(\mathfrak{A})$, we have only to find a maximal subset $\bar{\Delta}$ of $\Delta(\mathfrak{A})$ such that $f^{(i,j)}(x, y)$, $(i, j) \in \bar{\Delta}$, are linearly independent. Hence, for $(i, j) \in \partial\bar{\Delta}$, we obtain

$$h^{(i,j)}(x, y) = x^i y^j - g^{(i,j)}(x, y) \in \mathfrak{A}:\mathfrak{L},$$

where $g^{(i,j)}(x, y) \in \boldsymbol{P}(\bar{\Delta})$. In case of a nonprincipal ideal $\mathfrak{L} = (g^{(1)}, \cdots, g^{(L)})$ we may invoke Remark 3 by virtue of $\mathfrak{A}:\mathfrak{L} = \cap_{l=1}^{L}(\mathfrak{A}:(g^{(l)}))$.

V. Nonsemisimple Binary TDC Codes

Some semisimple TDC codes have been studied by several authors [2], [8], [9]. On the other hand, the error-correcting capabilities of nonsemisimple TDC codes with finite areas have never been investigated except for the radical codes [1] defined by check ideals that are equal to powers of the prime ideal $(x - 1, y - 1)$. A nonsemisimple code cannot be characterized solely by the zeros of the ideal, although any semisimple code can be determined by the zeros of the check or the generator ideal. For any TDC code C with area $m \times n$, the check ideal $\mathfrak{A}$ of C can be written in the form $\mathfrak{A} = (x^m - 1, f^{(1)}(x, y), \cdots, f^{(M)}(x, y), y^n - 1)$ for a certain system of polynomials $f^{(1)}, \cdots, f^{(M)}$. It is quite obvious that there exist a finite number of distinct ideals $\mathfrak{A} \supset \mathfrak{J}_{m,n}$ for given m, n. In order to enumerate all distinct TDC codes with the same area, we may judge which ideals coincide with one another by applying the normalization algorithm, by means of which we have explored nonsemisimple codes with small area for the first time. As any nonsemisimple TDC code is a direct sum of quasi-irreducible codes, we must first explore quasi-irreducible codes.

Example 3: A binary code $C_{\mathfrak{A}}$ with area 2×12 coincides with $C_{\mathfrak{A}_1} + C_{\mathfrak{A}_2}$, where $\mathfrak{A} = \mathfrak{A}_1 \cdot \mathfrak{A}_2$, $\mathfrak{A}_1 \supset (x^2 + 1, y^4 + 1)$ is a primary ideal belonging to the prime ideal $\mathfrak{P}_1 = (x + 1, y + 1)$ and $\mathfrak{A}_2 \supset (x^2 + 1, y^8 + y^4 + 1)$ is a primary ideal belonging to the prime ideal $\mathfrak{P}_2 = (x + 1, y^2 + y + 1)$. The zeros of these ideals are $V(\mathfrak{A}_1) = V(\mathfrak{P}_1) = \{(1, 1)\}$, $V(\mathfrak{A}_2) = V(\mathfrak{P}_2) = \{(1, \alpha), (1, \alpha^2)\}$, where $\alpha \in \mathrm{GF}(2^2)$ is a root of the minimum polynomial $x^2 + x + 1$. Such a former code $C_{\mathfrak{A}_1}$ with area 2×12 is nothing but a repetition of a code with area 2×4. In order to find all binary TDC codes with area 2×12, we have only to search out quasi-irreducible codes $C_{\mathfrak{Q}}$ defined by check ideals as follows

$$\mathfrak{Q}(= \mathfrak{A}_2) = \left(x^2 + 1, f^{(1)}(x, y), \cdots, f^{(M)}(x, y), y^8 + y^4 + 1\right), \tag{16}$$

where $f^{(k)}(1, \alpha) = 0$ and $\deg_x f^{(k)} \leq 1$, $\deg_y f^{(k)} \leq 7$, $1 \leq k \leq M$.

By applying the normalization algorithm to each of such ideal bases as in (16), we have obtained all distinct binary quasi-irreducible codes $C_{\mathfrak{Q}}$ with areas $n_1 \times n_2 = 2 \times 2$, 2×4, 2×6, 2×8, 2×10, 2×12, 4×4, 4×6, respectively. In Table I the check ideals (or check polynomials) are listed for several typical codes and some other codes related to them, the normal basis of each ideal being shown. Among the latter codes there are TDC codes with greater areas 6×6, 4×12, and others. The code $C_{\mathfrak{A}}$ with area 2×2, $\mathfrak{A} = (x^2 + 1, (x + 1)(y + 1), y^2 + 1)$, is a radical code [1]. The punctured TDC code with area 4×6 (designated by p in Table I) is a new quasi-cyclic code which is obtained by deleting the right six columns of every codeword (4×12 matrix) of the TDC code $C_{\mathfrak{A}}$ with area 4×12, $\mathfrak{A} = ((x + 1)^3, y^2 + y + x)$. The TDC code $C_{\mathfrak{A}}$ with area 2×6 (designated by s in Table I) is the direct sum of two quasi-irreducible codes $C_{\mathfrak{A}_1}, C_{\mathfrak{A}_2}$ with area 2×6, where $\mathfrak{A} = \mathfrak{A}_1\mathfrak{A}_2$, $\mathfrak{A}_1 = (x^2 + 1, y^2 + 1)$, $\mathfrak{A}_2 = (x^2 + 1, y^2 + y + x)$. Furthermore, the latter codes $C_{\mathfrak{A}}, C_{\mathfrak{A}_2}$ are equivalent to the respective known quasi-cyclic codes [11]. Every code listed in Table I is a linear code with the codelength n $(= n_1 n_2)$, the number k of information symbols and the minimum distance d. Some are optimal in the sense that they have the largest d among the linear codes with the same n and k (designated by * in Table I) [5]. Each of the TDC codes with $[n, k, d] = [8, 4, 4]$, $[12, 6, 4]$, $[24, 8, 8]$ has the same minimum distance as a best quasi-cyclic code [3], [11]. Best cyclic codes (designated by c) also are contained in Table I for comparison. Most of them are repetition codes or interleaved codes of cyclic codes with smaller length. The weight enumerators of the corresponding codes in Table I are shown in Table II. For example, a weight enumerator $6^2 1^4$ implies that the corresponding code has six codewords with weight two and one codeword with weight four (plus the zero codeword).

VI. Conclusion

We have proposed an algorithm for determining the independend point set of DP arrays. It implies that the problem of encoding TDC codes has been completely solved. In addition we have given a method of encoding their dual codes. By systematically exploring nonsemisimple binary TDC codes with comparatively small areas with the aid of our algorithm, we have found several optimal linear codes and their previously unknown TDC structures.

It is also possible to extend our investigation to Abelian codes. However, there remain some open problems such as

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TABLE I
QUASI-IRREDUCIBLE TDC CODES AND SOME CODES RELATED TO THEM

n	n_1	n_2	k	d	check ideal
c_4			3	2*	$(x+1)^3$
4	2	2	3	2*	$x^2+1, (x+1)(y+1), y^2+1$
c_8			4	2	x^4+1
8	2	4	4	4*	$x^2+1, y^2+(x+1)y+1$
c_{12}			4	4	x^4+x^2+1
12	2	6	4	6*	x^2+1, y^2+y+x
c_{12}			6	4*	$(x^2+x+1)^3$
12	2	6	6	4*	$x^2+1, (x+1)(y^2+y+1), y^4+y^2+1$
c_{12}			8	2	$(x^2+1)(x^2+x+1)^3$
12	2	6	8	2	x^2+1, y^4+y^2+1
s_{12}	2	6	8	3*	$x^2+1, y^4+xy^3+(x+1)y^2+xy+x$
c_{16}			6	4	$(x+1)^6$
16	2	8	6	6*	x^2+1, y^3+y^2+xy+1
c_{16}			10	2	$(x+1)^{10}$
16	2	8	10	4*	$x^2+1, (x+1)(y+1)^3, (y+1)^7$
16	4	4	10	4*	$x^4+1, (x+1)(y^2+(x^2+1)y+x^2), y^4+1$
16	4	4	10	4*	$x^4+1, (x^2+1)(y^2+1), y^3+xy^2+(x^3+x^2+x)y+x^3+x+1$
16	4	4	10	4*	$x^4+1, (x+1)(y^2+(x+1)y+x), y^4+1$
c_{20}			8	4	$(x+1)^3(x^5+1)$
20	2	10	8	6	$x^2+1, y^4+y^3+y^2+y+x$
20	2	10	8	8*	$x^2+1, y^4+y^3+xy^2+y+1$
c_{24}			6	8	$(x^2+x+1)^3$
24	2	12	6	8	$x^2+1, (x+1)(y^2+y+1), y^4+y^2+x$
24	4	6	6	8	$(x+1)^3, y^2+y+x^2$
p_{24}	4	6	6	10*	$(x+1)^3, y^2+y+x$
c_{24}			8	6	$(x+1)^5(x^3+1)$
24	2	12	8	6	x^2+1, y^4+y^2+x
24	4	6	8	6	x^4+1, y^2+y+x^2
24	2	12	8	8	$x^2+1, y^4+(x+1)y^3+y^2+(x+1)y+1$
24	4	6	8	8	$x^4+1, y^2+(x^3+x^2+x)y+1$
24	2	12	8	8	$x^2+1, y^4+y^2+(x+1)y+1$
24	4	6	8	8	x^4+1, y^2+y+x^3+x+1
24	2	12	8	8	$x^2+1, y^4+y^2+(x+1)y+x$
24	4	6	8	8	$x^4+1, y^2+y+x^3+x^2+x$
c_{24}			10	4	$(x^2+x+1)^5$
24	2	12	10	6	$x^2+1, (x+1)(y^4+y^2+1), y^6+y^5+y^3+(x+1)y^2+xy+x$
24	4	6	10	6	$x^4+1, (x+1)(y^2+y+x^2), y^4+y^2+1$
24	2	12	10	6	$x^2+1, (x+1)(y^4+y^2+1), y^6+y^5+y^3+y+x$
24	2	12	10	6	$x^2+1, (x+1)(y^4+y^2+1), y^6+y^5+y^3+xy+1$
36	6	6	4	18*	$x^4+x^2+1, y+x+1$
48	4	12	6	24*	$(x+1)^3, y^2+y+x$

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TABLE II
WEIGHT ENUMERATORS OF CORRESPONDING CODES IN TABLE I

n	n_1	n_2	k	d	weight enumerator
c_4			3	2*	$6^2 1^4$
4	2	2	3	2*	$6^2 1^4$
c_8			4	2	$4^2 6^4 4^6 1^8$
8	2	4	4	4*	$14^4 1^8$
c_{12}			4	4	$6^4 9^8$
12	2	6	4	6*	$12^6 3^8$
c_{12}			6	4*	$18^4 24^6 21^8$
12	2	6	6	4*	$18^4 24^6 21^8$
c_{12}			8	2	$6^2 63^4 116^6 63^8 6^{10} 1^{12}$
12	2	6	8	2	$12^2 54^4 108^6 81^8$
s_{12}	2	6	8	3*	$16^3 39^4 48^5 48^6 48^7 39^8 16^9 1^{12}$
c_{16}			6	4	$12^4 38^8 12^{12} 1^{16}$
16	2	8	6	6*	$16^6 30^8 16^{10} 1^{16}$
c_{16}			10	2	$8^2 60^4 184^6 518^8 184^{10} 60^{12} 8^{14} 1^{16}$
16	2	8	10	4*	$76^4 192^6 486^8 192^{10} 76^{12} 1^{16}$
16	4	4	10	4*	$76^4 192^6 486^8 192^{10} 76^{12} 1^{16}$
16	4	4	10	4*	$68^4 224^6 438^8 224^{10} 68^{12} 1^{16}$
16	4	4	10	4*	$60^4 256^6 390^8 256^{10} 60^{12} 1^{16}$
c_{20}			8	4	$5^4 4^5 20^7 10^8 40^9 96^{10} 40^{11} 10^{12} 20^{13} 4^{15} 5^{16} 1^{20}$
20	2	10	8	6	$20^6 50^8 120^{10} 40^{12} 20^{14} 5^{16}$
20	2	10	8	8*	$130^8 120^{12} 5^{16}$
c_{24}			6	8	$18^8 24^{12} 21^{16}$
24	2	12	6	8	$6^8 48^{12} 9^{16}$
24	4	6	6	8	$6^8 48^{12} 9^{16}$
p_{24}	4	6	6	10*	$24^{10} 12^{12} 24^{14} 3^{16}$
c_{24}			8	6	$12^6 3^8 36^{10} 152^{12} 36^{14} 3^{16} 12^{18} 1^{24}$
24	2	12	8	6	$24^6 6^8 144^{12} 72^{14} 9^{16}$
24	4	6	8	6	$24^6 6^8 144^{12} 72^{14} 9^{16}$
24	2	12	8	8	$42^8 168^{12} 45^{16}$
24	4	6	8	8	$42^8 168^{12} 45^{16}$
24	2	12	8	8	$30^8 48^{10} 96^{12} 48^{14} 33^{16}$
24	4	6	8	8	$30^8 48^{10} 96^{12} 48^{14} 33^{16}$
24	2	12	8	8	$18^8 96^{10} 24^{12} 96^{14} 21^{16}$
24	4	6	8	8	$18^8 96^{10} 24^{12} 96^{14} 21^{16}$
c_{24}			10	4	$12^4 102^8 192^{10} 396^{12} 192^{14} 129^{16}$
24	2	12	10	6	$24^6 114^8 144^{10} 408^{12} 216^{14} 117^{16}$
24	4	6	10	6	$24^6 114^8 144^{10} 408^{12} 216^{14} 117^{16}$
24	2	12	10	6	$8^6 114^8 192^{10} 408^{12} 168^{14} 117^{16} 16^{18}$
24	2	12	10	6	$14^6 78^8 282^{10} 288^{12} 258^{14} 81^{16} 22^{18}$
36	6	6	4	18*	$12^{18} 3^{24}$
48	4	12	6	24*	$60^{24} 3^{32}$

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those of constructing good TDC and Abelian codes more efficiently and of decoding them.

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A Method for Proving Multiterminal Source Coding Theorems

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Abstract—**The two-step method used by Wyner and Ziv to prove the Wyner–Ziv theorem is extended to prove sliding-block source coding theorems for coding a general finite-alphabet ergodic multiterminal source with respect to a single-letter fidelity criterion. The first step replaces every stochastic encoding in the network by a deterministic sliding-block encoding. The second step involves using the Slepian–Wolf theorem to adjust the sliding-block encodings so that they will have the desired rate, while introducing little additional distortion. The method is applied to give quick proofs of sliding-block versions of theorems of Berger, Kaspi, and Tung. Since the method obtains sliding-block coders directly without first obtaining block coders, the block coding results can be obtained as an easy corollary. The direct methods for proving results on block coding are more difficult and do not imply the corresponding results on sliding-block coding. Indeed, in the multiterminal case it is, in general, not known how to construct good sliding-block coders from good block coders.**

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LET X, Y be jointly stationary and ergodic processes. Let the stationary process Z be a stochastic encoding of X. Suppose some block decoding of (Z, Y) yields an estimate of X to within the distortion level D. The Wyner–Ziv theorem says that if $R > I(X, Z \mid Y)$, then a block encoding $\hat{Z}$ of X at a rate less than R can be found so that $(\hat{Z}, Y)$ yields an estimate of X within the level D by means of an appropriate block decoding. Wyner and Ziv prove their result [11] in two steps. First, they find a block encoding U of X so that $H(U \mid Y) = I(X, U \mid Y) < R$ and (U, Y) yields X to within D. (That is, U maintains the same information rate inequality and distortion level as Z.) The second step is to apply the Slepian–Wolf theorem to the pair process (U, Y) to get a block encoding $\hat{Z}$ of U at rate less than R so that a block decoding of $(\hat{Z}, Y)$ gives such a close estimate to U that $(\hat{Z}, Y)$ can yield an estimate of X to within D. The process $\hat{Z}$ is the desired block encoding of X.

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